

simple-plot

Typst Package

Mathematical Function Plotting

A lightweight library for creating elegant mathematical plots

Version 0.9.1

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1. Introduction

`simple-plot` is a Typst package for creating clean, elegant mathematical plots. Built on CeTZ, it provides an intuitive interface for plotting functions, data points, and creating publication-ready graphs.

1.1. Features

- Plot mathematical functions with automatic sampling
- Parametric curves (ellipses, hyperbolas, closed shapes)
- Scatter plots and line plots with customizable markers
- Fill areas under curves or between two curves (solid or hatched)
- Clean integer-based tick system by default
- Major and minor grid with elegant styling
- Gap-based grid line breaks around tick labels (`grid-label-break`)
- Automatic axis extension beyond grid
- Flexible axis positioning (origin, bottom/left, custom)
- Multiple label display options (`unit-label-only`, `label-step`)
- Function labels with flexible positioning
- Text annotations and reference lines (`vline`, `hline`)
- Riemann sum rectangles with left, right, midpoint, lower, and upper methods
- Volume of revolution diagrams with end caps, disks, and tilted axes
- Clipping for clean rendering at boundaries
- Global defaults and style overrides

1.2. Installation

Import the package in your Typst document:

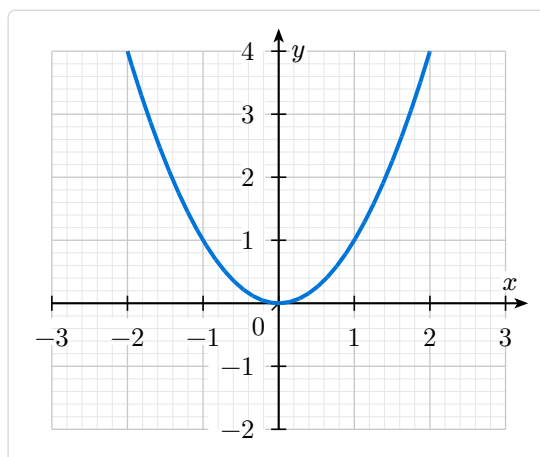
```
#import "@preview/simple-plot:1.0.0": plot
```

1.3. Quick Start

Code

```
#plot(  
  width: 6, height: 5,  
  xmin: -3, xmax: 3, ymin: -2, ymax: 4,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: true,  
  (fn: x => x * x, stroke: blue + 1.5pt),  
)
```

Preview



2. Basic Usage

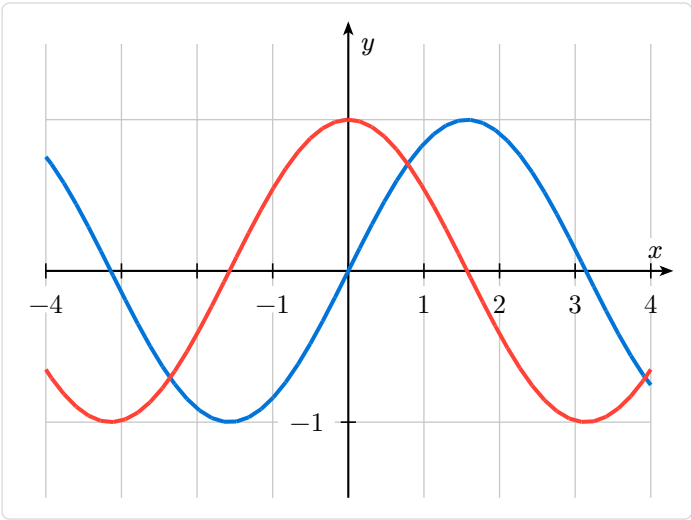
2.1. Plotting Functions

Plot mathematical functions by passing a dictionary with `fn`:

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: -4, xmax: 4, ymin: -1.5, ymax: 1.5,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "major",  
  (fn: x => calc.sin(x), stroke: blue + 1.5pt),  
  (fn: x => calc.cos(x), stroke: red + 1.5pt),  
)
```

Preview



2.2. Mathematical Functions Reference

Functions are defined using Typst's `calc` module. Here are the most common mathematical functions:

Function	Typst syntax
Power x^n	<code>calc.pow(x, n)</code>
Square root $\sqrt{x}$	<code>calc.sqrt(x)</code>
Absolute value $ x $	<code>calc.abs(x)</code>
Sine $\sin(x)$	<code>calc.sin(x)</code>
Cosine $\cos(x)$	<code>calc.cos(x)</code>
Tangent $\tan(x)$	<code>calc.tan(x)</code>
Exponential e^x	<code>calc.exp(x)</code>
Natural log $\ln(x)$	<code>calc.ln(x)</code>
Log base b	<code>calc.log(x, base: b)</code>
Maximum	<code>calc.max(a, b)</code>
Minimum	<code>calc.min(a, b)</code>

Important: When using constants in calculations, use decimal notation (e.g., 2.0 instead of 2) to avoid type errors. For example:

- ✓ $x \Rightarrow x * x / 2.0$
- ✗ $x \Rightarrow x * x / 2$ (*may cause errors*)

This is because Typst's type system requires consistent float arithmetic.

To draw a function with a hole or discontinuity, return `none` from the function at that point — the line will break without connecting across the gap:

```
// Draws 1/x with a break at x=0
(fn: x => if calc.abs(x) < 0.01 { none } else { 1.0 / x }, stroke: blue + 1.5pt)
```

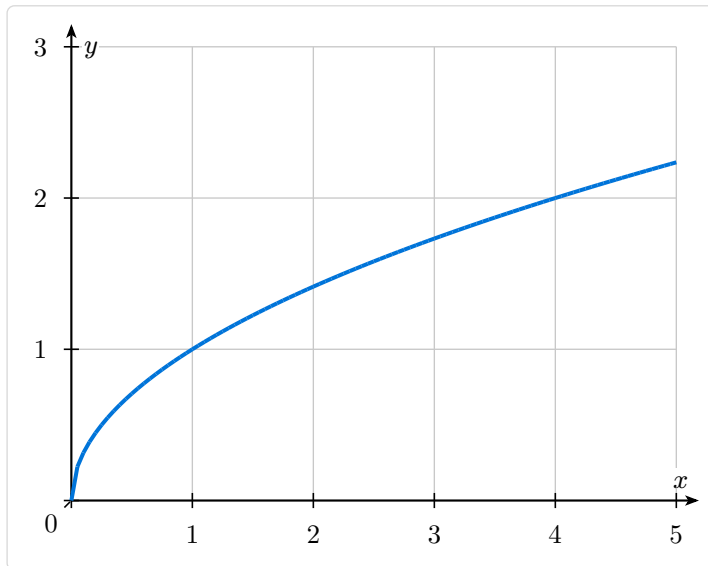
2.3. Function Domain

Specify a custom domain for functions:

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 5, ymin: 0, ymax: 3,  
  xlabel: "$x$", ylabel: "$y$",  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  show-grid: "major",  
  (fn: x => calc.sqrt(x), domain: (0, 5), stroke: blue + 1.5pt),  
)
```

Preview



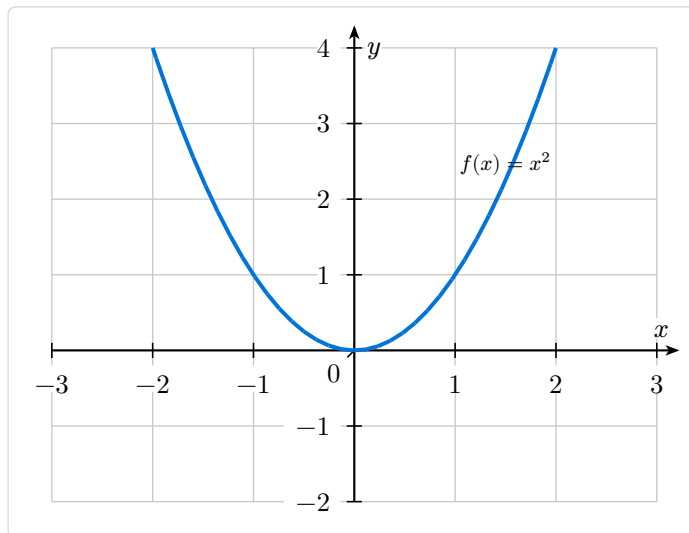
2.4. Function Labels

Add a `label:` field to any function spec. By default, labels are placed **automatically**: the engine scores candidate positions along the curve and around each point, avoiding other curves, tick labels, axes, and previously placed labels. Set `label-anchor` or `label-side` to override placement.

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: -3, xmax: 3, ymin: -2, ymax: 4,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "major",  
  (  
    fn: x => x * x,  
    stroke: blue + 1.5pt,  
    label:  $f(x) = x^2$ ,  
    label-side: "above",  
    label-pos: 0.75,  
  ),  
)
```

Preview



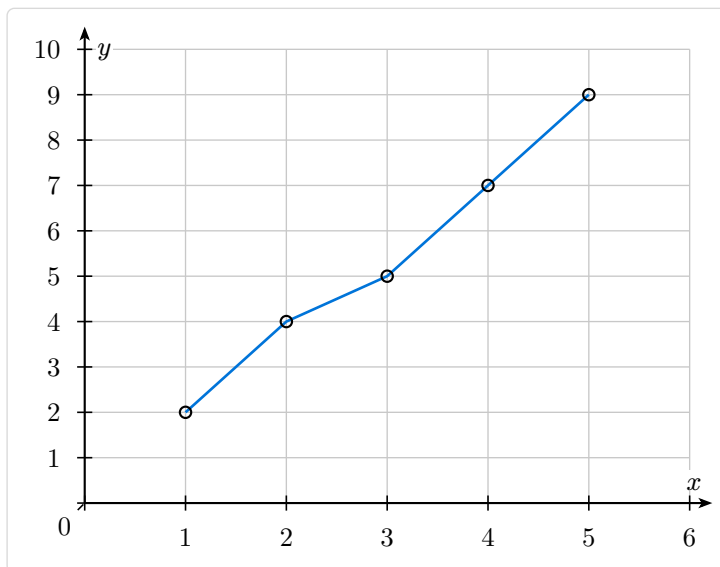
2.5. Data Points and Scatter Plots

Plot discrete data points:

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 6, ymin: 0, ymax: 10,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  show-grid: "major",  
  (  
    data: ((1, 2), (2, 4), (3, 5), (4, 7), (5, 9)),  
    mark: "o",  
    mark-size: 0.15,  
    stroke: blue + 1pt,  
  ),  
)
```

Preview



3. Grid Options

3.1. Grid Modes

Control grid display with `show-grid`:

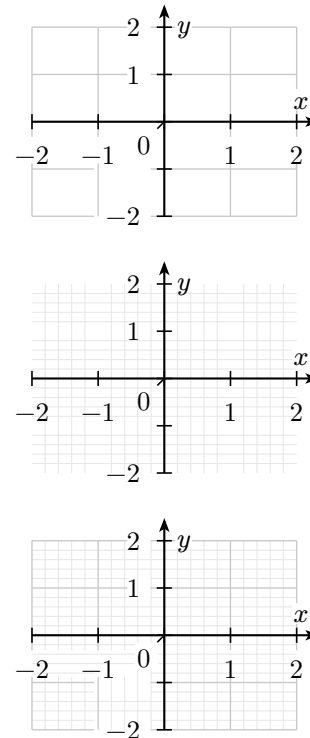
Code

```
// Major grid only
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "major",
)

// Minor grid only
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "minor",
)

// Both grids
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "both",
)
```

Preview



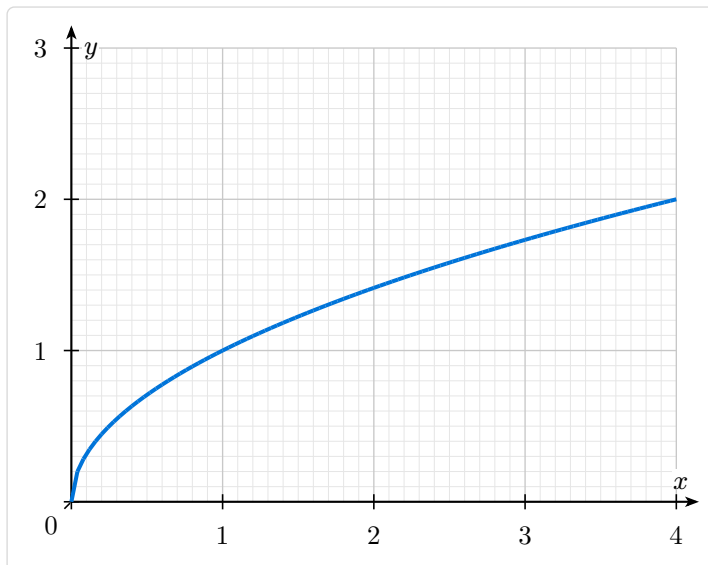
3.2. Minor Grid Subdivisions

Control the number of subdivisions with `minor-grid-step`:

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 4, ymin: 0, ymax: 3,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  show-grid: "both",  
  minor-grid-step: 10, // 10 subdivisions per unit  
  (fn: x => calc.sqrt(x), domain: (0, 4), stroke: blue + 1.5pt),  
)
```

Preview



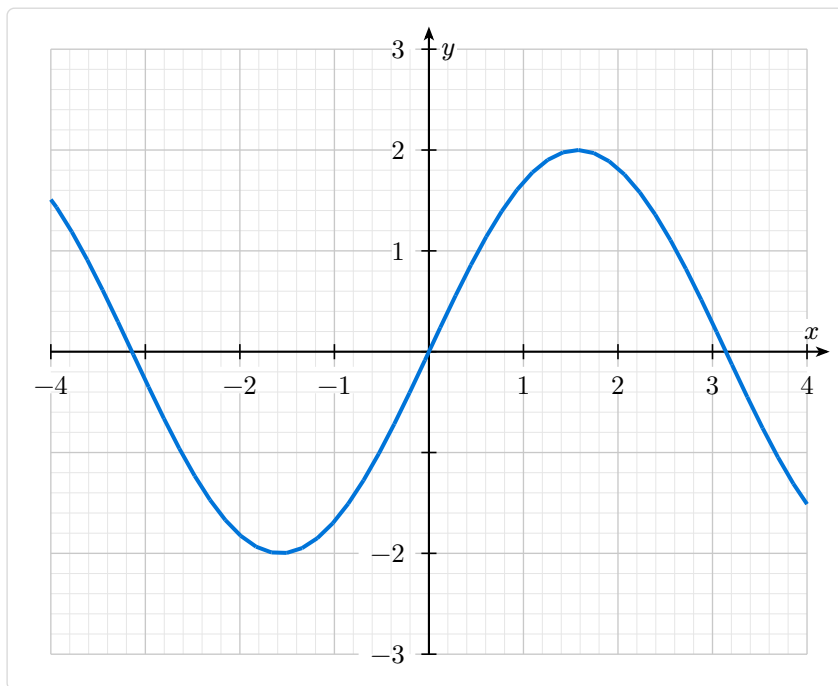
3.3. Grid Label Break

The `grid-label-break` option (enabled by default) draws grid lines with gaps around tick labels, creating an elegant break effect. Unlike a white-box approach, this works on any background color:

Code

```
#plot(  
  width: 10, height: 8,  
  xmin: -4, xmax: 4, ymin: -3, ymax: 3,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "both",  
  minor-grid-step: 5,  
  grid-label-break: true, // Default  
  (fn: x => calc.sin(x) * 2, stroke: blue + 1.5pt),  
)
```

Preview



4. Axis Configuration

4.1. Axis Position

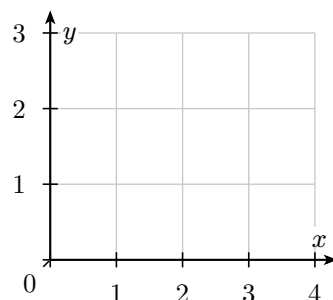
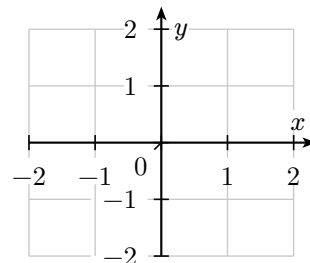
Position axes at origin (default), bottom/left, or custom values:

Code

```
// Through origin (default)
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "major",
)

// Bottom and left
#plot(
  width: 5, height: 4,
  xmin: 0, xmax: 4, ymin: 0, ymax: 3,
  axis-x-pos: "bottom",
  axis-y-pos: "left",
  show-grid: "major",
)
```

Preview



4.2. Axis Extension

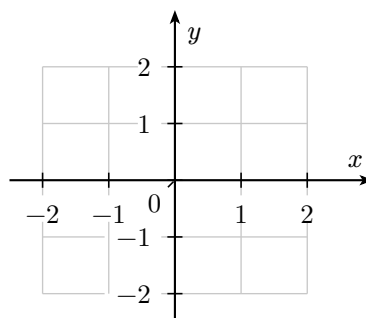
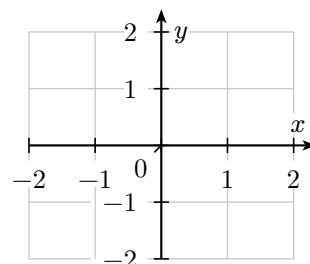
By default, axes extend 0.3 cm beyond the grid on the arrow side, independent of the data scale. Customize with `axis-x-extend` and `axis-y-extend` — a length (0.5cm, 3mm) is absolute, a bare number is interpreted in data units (legacy):

Code

```
// Default extension (0, 0.5)
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "major",
)

// Custom extension
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  axis-x-extend: (0.5, 1),
  axis-y-extend: (0.5, 1),
  show-grid: "major",
)
```

Preview



5. Tick Configuration

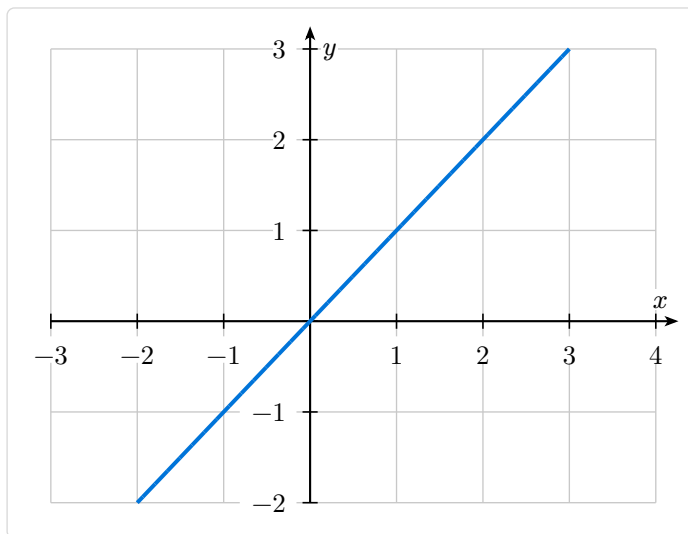
5.1. Default Integer Ticks

By default, ticks are placed at every integer (step = 1):

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: -3, xmax: 4, ymin: -2, ymax: 3,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "major",  
  (fn: x => x, stroke: blue + 1.5pt),  
)
```

Preview



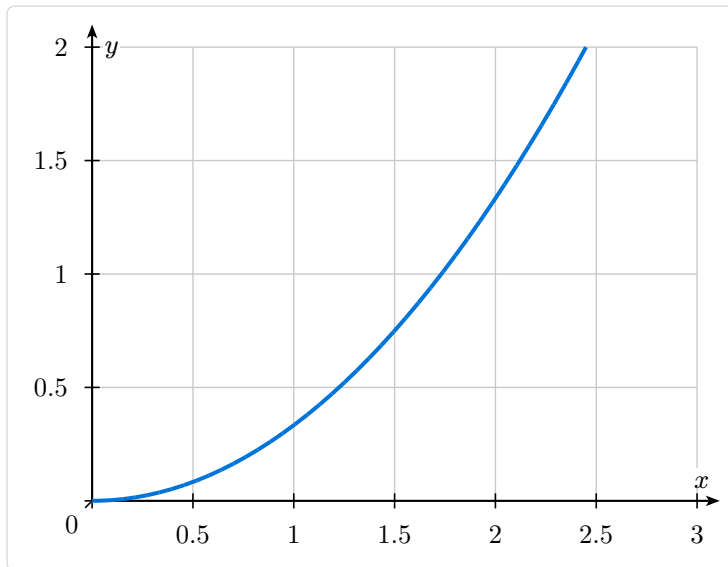
5.2. Custom Tick Step

Change tick spacing with `xtick-step` and `ytick-step`:

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 3, ymin: 0, ymax: 2,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  xtick-step: 0.5,  
  ytick-step: 0.5,  
  show-grid: "major",  
  (fn: x => x * x / 3, stroke: blue + 1.5pt),  
)
```

Preview



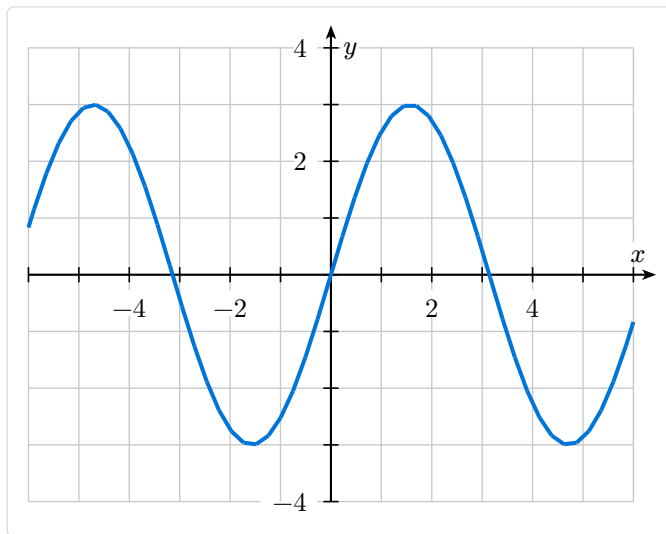
5.3. Tick Label Step

Show labels only at every N-th tick with `xtick-label-step` and `ytick-label-step`:

Code

```
#plot(
  width: 8, height: 6,
  xmin: -6, xmax: 6, ymin: -4, ymax: 4,
  xlabel:  $x$ , ylabel:  $y$ ,
  xtick-label-step: 2, // Labels at -6, -4, -2, 2, 4, 6
  ytick-label-step: 2, // Labels at -4, -2, 2, 4
  show-grid: "major",
  (fn: x => calc.sin(x) * 3, stroke: blue + 1.5pt),
)
```

Preview



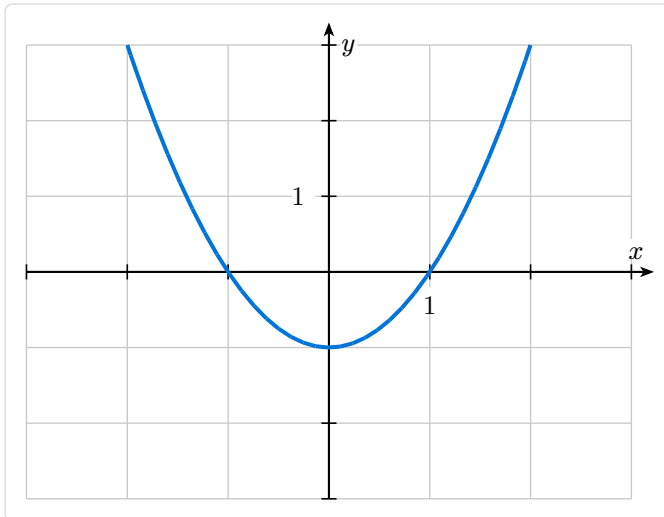
5.4. Unit Label Only

Show only “1” on each axis for a minimal style:

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: -3, xmax: 3, ymin: -3, ymax: 3,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  unit-label-only: true,  
  show-origin: false,  
  show-grid: "major",  
  (fn: x => x * x - 1, stroke: blue + 1.5pt),  
)
```

Preview



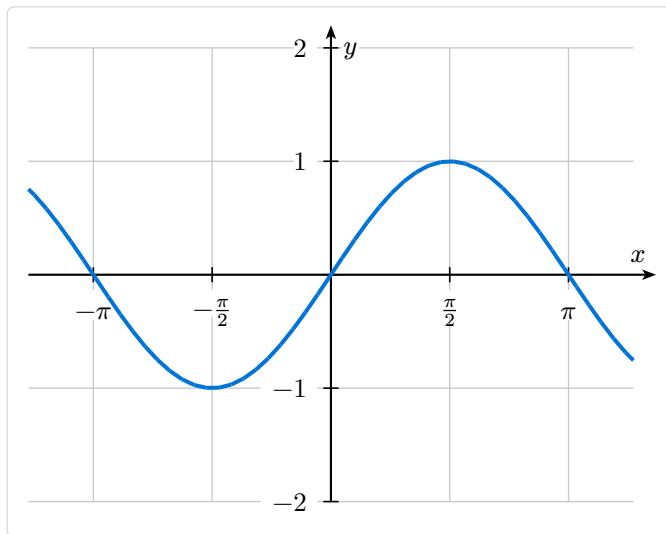
5.5. Custom Tick Positions

Specify exact tick positions with `xtick` and `ytick`:

Code

```
#plot(
  width: 8, height: 6,
  xmin: -4, xmax: 4, ymin: -2, ymax: 2,
  xlabel:  $x$ , ylabel:  $y$ ,
  xtick: (-calc.pi, -calc.pi/2, 0, calc.pi/2, calc.pi),
  xtick-labels: ( $-\pi$ ,  $-\pi/2$ ,  $0$ ,  $\pi/2$ ,  $\pi$ ),
  show-grid: "major",
  (fn: x => calc.sin(x), stroke: blue + 1.5pt),
)
```

Preview



5.6. Hide Origin Label

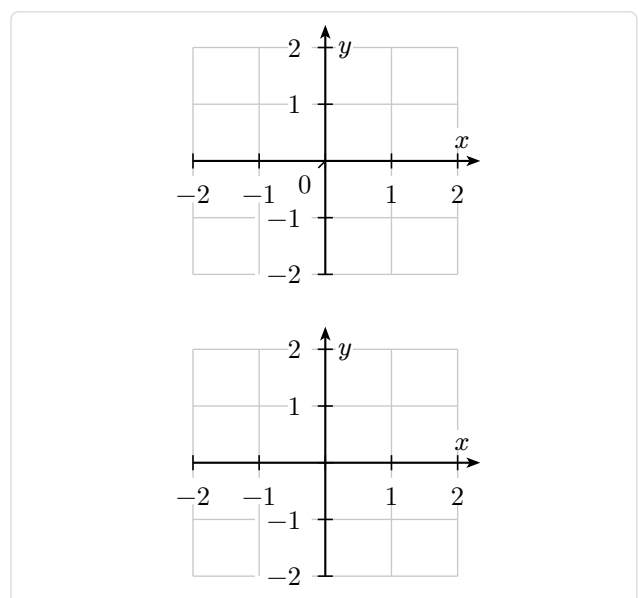
Control the “0” label at the origin:

Code

```
// With origin (default)
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-origin: true,
  show-grid: "major",
)

// Without origin
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-origin: false,
  show-grid: "major",
)
```

Preview



6. Markers

6.1. Available Marker Types

The following markers are available:

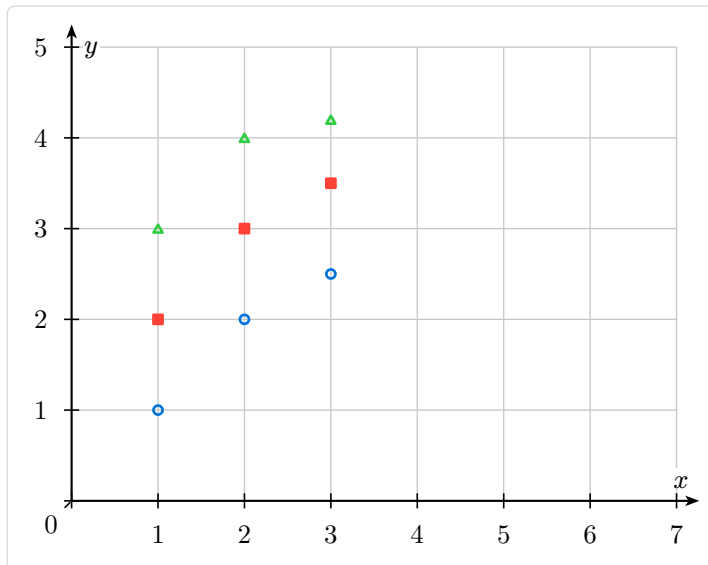
Marker	Description
"o"	Circle (outline)
"*"	Circle (filled)
"square"	Square (outline)
"square*"	Square (filled)
"triangle"	Triangle up (outline)
"triangle*"	Triangle up (filled)
"diamond"	Diamond (outline)
"diamond*"	Diamond (filled)
"star"	Star (outline)
"star*"	Star (filled)
"+"	Plus sign
"x"	Cross
" "	Vertical bar
"_"	Horizontal bar

6.2. Using Markers

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 7, ymin: 0, ymax: 5,  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  show-grid: "major",  
  data(((1, 1), (2, 2), (3, 2.5))), mark: "o", mark-stroke: blue),  
  data(((1, 2), (2, 3), (3, 3.5))), mark: "square*", mark-fill: red, mark-stroke: red),  
  data(((1, 3), (2, 4), (3, 4.2))), mark: "triangle", mark-stroke: green),  
)
```

Preview



7. Convenience Functions

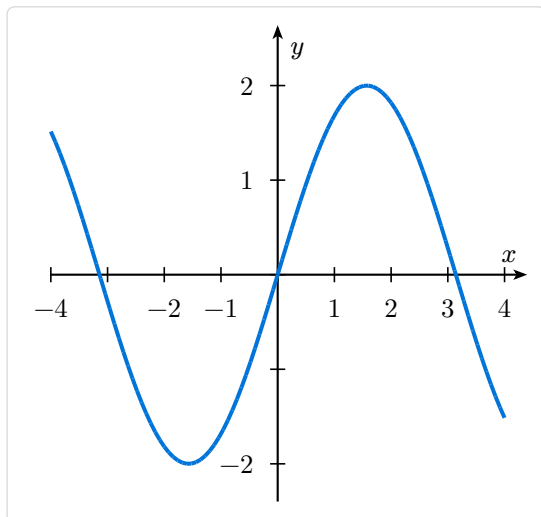
7.1. plot-fn - Quick Function Plot

Plot a single function with automatic y-scaling:

Code

```
#plot-fn(  
  x => calc.sin(x) * 2.0,  
  domain: (-4, 4),  
  stroke: blue + 1.5pt,  
)
```

Preview



Parameters:

Parameter	Type	Default	Description
fn	function	—	Function to plot
domain	array	(-5, 5)	X domain
ymin / ymax	float/auto	auto	Y bounds; auto = computed from samples
stroke	stroke	blue + 1.2pt	Line style
samples	int	100	Sample count (auto y-range and curve)
..args	any	—	Forwarded to plot()

7.2. plot-rational - Rational Function Wrapper

plot-rational is a convenience wrapper for common rational-function exercise graphs. It draws a single function on a centered coordinate grid and forwards extra arguments to plot, so you can add horizontal asymptotes, holes, or labels. Pass vertical asymptotes with vertical-asymptotes; the function domain is split around those values to avoid sampling at the pole.

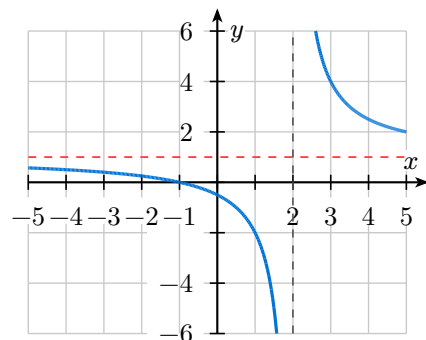
```
#plot-rational(  
  x => (x + 1) / (x - 2),  
  xmin: -5, xmax: 5,  
  ymin: -6, ymax: 6,  
  vertical-asymptotes: (2,),
```

```
hline(1, stroke: stroke(paint: red, thickness: 0.7pt, dash: "dashed"),
)
```

Code

```
#plot-rational(
  x => (x + 1) / (x - 2),
  xmin: -5, xmax: 5,
  ymin: -6, ymax: 6,
  vertical-asymptotes: (2,),
  hline(1, stroke: stroke(paint: red, thickness: 0.7pt,
dash: "dashed")),
)
```

Preview



Parameters:

Parameter	Type	Default	Description
fn	function	—	Function to plot
xmin / xmax	float	-6 / 6	X-axis bounds
ymin / ymax	float	-8 / 8	Y-axis bounds
stroke	stroke	blue + 1.2pt	Function stroke
domain	array/auto	auto	Function domain; auto = (xmin, xmax)
samples	int	200	Function sample count
vertical-asymptotes	array	()	X-values to skip and draw as vertical dashed lines
asymptote-stroke	stroke	dashed gray	Stroke for vertical asymptote lines
asymptote-gap	float	0.05	Half-gap removed around each vertical asymptote
width / height	float/auto	auto	Plot dimensions
..args	any	—	Forwarded to plot(); positional args become extra series

7.3. limit-schema / schema-lim - Local Limit Sketches

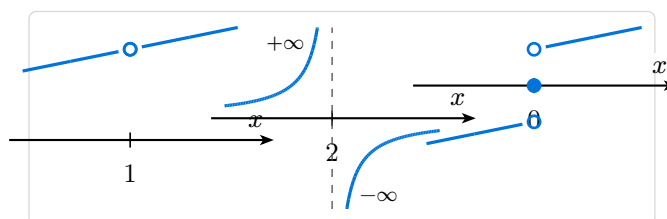
limit-schema draws a small schematic graph around a point **a**. It is useful for removable discontinuities, jumps, and vertical asymptotes. schema-lim is kept as a French alias.

```
#limit-schema(a: 1, left: 4, right: 4)
#limit-schema(a: 2, left: "+oo", right: "-oo")
#limit-schema(a: 0, left: -1, right: 1, val: 0)
```

Code

```
#grid(columns: (1fr, 1fr, 1fr), gutter: 0.5em,
  limit-schema(a: 1, left: 4, right: 4),
  limit-schema(a: 2, left: "+oo", right: "-oo"),
  limit-schema(a: 0, left: -1, right: 1, val: 0),
)
```

Preview



Parameters:

Parameter	Type	Default	Description
a	float	0	X-position of the local behavior
left	float/string/ none	none	Left-hand behavior: finite value, "+oo", "-oo", or none
right	float/string/ none	none	Right-hand behavior: finite value, "+oo", "-oo", or none
val	float/none	none	Defined value $f(a)$ as a filled dot
a-label	content/auto	auto	Tick label at a; auto prints the value
show-limit	bool/float	false	Draw a dashed horizontal limit line; true auto-detects a common finite limit
l-label	content/auto	auto	Label for the limit line
width / height	float	3.8 / 2.8	Schema dimensions in cm

7.4. scatter / data - Point Sets

`scatter` and `data` are identical functions. Use them to create isolated point specifications. Set `connect: true` to join points with a line.

```
#plot(
  xmin: 0, xmax: 4, ymin: 0, ymax: 6,
  scatter(((1, 2), (2, 4), (3, 5))), mark: "o", mark-fill: blue),
  data(((1, 1.5), (2, 3.5), (3, 4.5))), mark: "*", mark-fill: red),
)
```

Parameters (both functions):

Parameter	Type	Default	Description
points	array	—	Array of (x, y) tuples
mark	string	"*"	Marker type
mark-size	float	0.12	Marker size in cm
mark-fill	color	blue	Marker fill color
mark-stroke	stroke	blue + 0.8pt	Marker stroke
connect	bool	false	Join points with a line
stroke	stroke	none	Line stroke when <code>connect: true</code>
label	content	none	Series label
label-pos	float	0.8	Position along series in [0, 1]

label-anchor	string	“south-west”	Label anchor
--------------	--------	--------------	--------------

7.5. line-plot - Connected Line Plot

Create a line plot with markers (points joined by default):

```
#plot(
  xmin: 0, xmax: 4, ymin: 0, ymax: 6,
  line-plot(((1, 2), (2, 4), (3, 5))), stroke: blue + 1pt, mark: "o"),
)
```

Parameters:

Parameter	Type	Default	Description
points	array	—	Array of (x, y) tuples
stroke	stroke	blue + 1.2pt	Line stroke
mark	string	"o"	Marker type
mark-size	float	0.1	Marker size in cm
mark-fill	color	white	Marker fill
mark-stroke	stroke	blue + 0.8pt	Marker stroke
label	content	none	Series label
label-pos	float	0.8	Position along series in [0, 1]
label-anchor	string	“south-west”	Label anchor

7.6. func-plot - Function Plot Helper

Build a function series spec with full marker and label control:

```
#let my-func = func-plot(
  x => calc.sin(x),
  stroke: blue + 1.5pt,
  label: $sin(x)$,
  mark: "o",
  mark-interval: 15, // marker every 15th sample
)
#plot(xmin: -4, xmax: 4, ymin: -1.5, ymax: 1.5, my-func)
```

Parameters:

Parameter	Type	Default	Description
fn	function	—	Function to plot (positional or fn:)
domain	array/auto	auto	X domain; auto = full plot range
stroke	stroke	blue + 1.2pt	Line style
samples	int	100	Sample count
mark	string	"none"	Marker type
mark-size	float	0.1	Marker size in cm
mark-fill	color	blue	Marker fill
mark-stroke	stroke	blue + 0.8pt	Marker stroke

mark-interval	int	10	Draw a marker every N-th sample
label	content	none	Curve label
label-pos	float	0.8	Position in [0, 1]
label-anchor	string	“south-west”	Label anchor

7.7. parametric - Ellipses and Hyperbolas

Draw curves that are not single-valued functions of x , such as ellipses:

```
#plot(
  xmin: -3, xmax: 3, ymin: -2, ymax: 2,
  parametric(t => 2 * calc.cos(t), t => calc.sin(t), domain: (0, 2 * calc.pi)),
)
```

For a hyperbola branch, use a finite parameter interval:

```
#plot(
  xmin: -4, xmax: 4, ymin: -3, ymax: 3,
  parametric(t => (calc.exp(t) + calc.exp(-t)) / 2, t => (calc.exp(t) - calc.exp(-t)) / 2,
  domain: (-1.5, 1.5)),
  parametric(t => -(calc.exp(t) + calc.exp(-t)) / 2, t => (calc.exp(t) - calc.exp(-t)) / 2,
  domain: (-1.5, 1.5)),
)
```

Parameters:

Parameter	Type	Default	Description
fn-x	function	—	X coordinate as function of t
fn-y	function	—	Y coordinate as function of t
domain	array	(0.0, 1.0)	Parameter range (t_1, t_2)
stroke	stroke	blue + 1.2pt	Curve stroke
samples	int	100	Number of parameter samples

7.8. fill-area - Fill Under a Curve

Fill the region between a function and a baseline (default: $y = 0$):

```
#plot(
  xmin: 0, xmax: calc.pi, ymin: -0.2, ymax: 1.2,
  fill-area(x => calc.sin(x), domain: (0, calc.pi), color: blue.lighten(70%)),
  (fn: x => calc.sin(x), domain: (0, calc.pi), stroke: blue + 1.5pt),
)
```

Use `baseline` for a non-zero base level, or `hatch` for hatched fills — either the flat form or a single dict:

```
fill-area(x => calc.sin(x), domain: (0, calc.pi),
  hatch: "ne", hatch-spacing: 5pt, hatch-stroke: blue + 0.5pt)
// equivalent:
fill-area(x => calc.sin(x), domain: (0, calc.pi),
  hatch: (style: "ne", spacing: 5pt, stroke: blue + 0.5pt))
```

Parameters:

Parameter	Type	Default	Description
fn	function	—	Function $f(x)$ bounding the top of the region (positional or fn:)
domain	array/auto	auto	X interval; auto = full plot range
baseline	float	0.0	Y-value of the bottom of the region
color	color	luma(220)	Fill color; none for hatch-only
hatch	str/dict/none	none	Hatch pattern: "ne", "nw", "h", "v", "cross", "grid" — or (style: "ne", spacing: 5pt, stroke: red + 1pt)
hatch-spacing	length	5pt	Spacing between hatch lines
hatch-stroke	stroke	luma(80) + 0.5pt	Hatch line stroke
samples	int	80	Sample count

7.9. area-between - Fill Between Two Curves

Fill the region enclosed by two functions:

```
#plot(
  xmin: -1, xmax: 3, ymin: -1, ymax: 10,
  area-between(x => x * x, x => 3 * x, domain: (0, 3),
    color: green.lighten(60%)),
  (fn: x => x * x, stroke: blue + 1.5pt),
  (fn: x => 3 * x, stroke: red + 1.5pt),
)
```

Parameters:

Parameter	Type	Default	Description
fn1	function	—	First bounding function (positional or fn1:)
fn2	function	—	Second bounding function (positional or fn2:)
domain	array/auto	auto	X interval; auto = full plot range
color	color	luma(220)	Fill color; none for hatch-only
hatch	str/dict/none	none	Hatch pattern (style string or (style: ..., spacing: ..., stroke: ...) dict)
hatch-spacing	length	5pt	Hatch line spacing
hatch-stroke	stroke	luma(80) + 0.5pt	Hatch line stroke
samples	int	80	Sample count

7.10. fill-closed - Fill a Parametric Closed Curve

Fill the interior of a closed parametric curve (the curve should start and end at the same point):

```
#plot(
  xmin: -3, xmax: 3, ymin: -2, ymax: 2,
  fill-closed(t => 2 * calc.cos(t), t => calc.sin(t),
    domain: (0, 2 * calc.pi), color: blue.lighten(70%)),
)
```

```
parametric(t => 2 * calc.cos(t), t => calc.sin(t),
           domain: (0, 2 * calc.pi), stroke: blue + 1.5pt),
)
```

Parameters:

Parameter	Type	Default	Description
fn-x	function	—	X coordinate as function of t
fn-y	function	—	Y coordinate as function of t
domain	array	(0.0, 1.0)	Parameter range; curve should be closed
color	color	luma(220)	Fill color; none for hatch-only
hatch	str/dict/none	none	Hatch pattern (style string or (style: .., spacing: .., stroke: ..) dict)
hatch-spacing	length	5pt	Hatch line spacing
hatch-stroke	stroke	luma(80) + 0.5pt	Hatch line stroke
samples	int	80	Sample count

7.11. vline / hline - Reference Lines

Draw vertical or horizontal reference lines at a fixed coordinate:

```
#plot(
  xmin: -3, xmax: 3, ymin: -2, ymax: 4,
  (fn: x => x * x - 1, stroke: blue + 1.5pt),
  vline(1.0, stroke: red + 0.8pt + (dash: "dashed")),
  hline(0.0, stroke: gray + 0.8pt),
)
```

Use ymin/ymax (for vline) or xmin/xmax (for hline) to draw partial lines:

```
vline(2.0, ymin: 0, ymax: 4)           // vertical from y=0 to y=4
hline(1.5, xmin: 0, xmax: 3)          // horizontal from x=0 to x=3
```

vline parameters:

Parameter	Type	Default	Description
x0	float	—	X position of the line
stroke	stroke	luma(100) + 0.6pt	Line stroke
ymin	float/auto	auto	Bottom y limit (default: plot ymin)
ymax	float/auto	auto	Top y limit (default: plot ymax)

hline parameters:

Parameter	Type	Default	Description
y0	float	—	Y position of the line
stroke	stroke	luma(100) + 0.6pt	Line stroke
xmin	float/auto	auto	Left x limit (default: plot xmin)

xmax	float/auto	auto	Right x limit (default: plot xmax)
------	------------	------	------------------------------------

7.12. note - Text Annotation

Place a text annotation at a data-coordinate position:

```
#plot(
  xmin: -3, xmax: 3, ymin: -2, ymax: 4,
  (fn: x => x * x - 1, stroke: blue + 1.5pt),
  note([f(x) = x^2 - 1], pos: (1.5, 2.5), anchor: "west", size: 9pt),
)
```

Parameters:

Parameter	Type	Default	Description
body	content	—	Annotation text or content
pos	array	—	(x, y) in data coordinates
anchor	string	“center”	Text anchor point
size	length	9pt	Font size

7.13. riemann-sum - Riemann Sum Rectangles

Draw left, right, midpoint, lower, or upper Riemann sum rectangles for a function. Use it as a plot item alongside the function curve or fill-area.

The five methods:

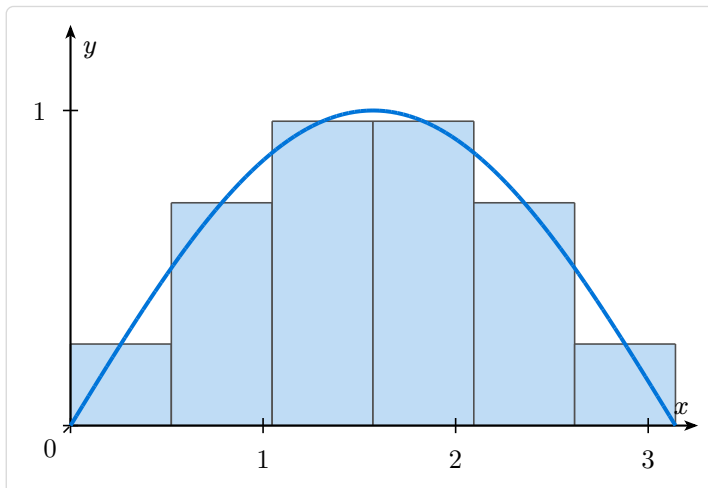
- "left" / "right" / "mid" — evaluation at the left endpoint, right endpoint, or midpoint of each sub-interval
- "lower" / "upper" — true infimum / supremum sampled within each sub-interval; correct for any function shape including U-curves

7.13.1. Basic Example

Code

```
#plot(
  width: 8, height: 5,
  xmin: 0, xmax: calc.pi, ymin: 0, ymax: 1.2,
  axis-x-pos: "bottom", axis-y-pos: "left",
  riemann-sum(
    x => calc.sin(x),
    domain: (0.0, calc.pi),
    n: 6, method: "mid",
    color: blue.lighten(75%),
  ),
  (fn: x => calc.sin(x), domain: (0.0, calc.pi), stroke: blue + 1.5pt),
)
```

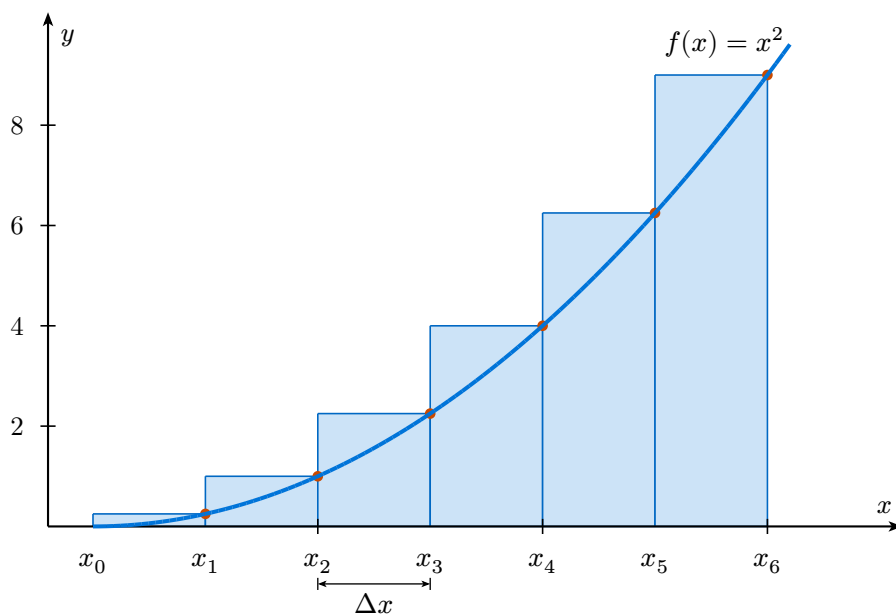
Preview



7.13.2. Annotations: show-points, show-dx, show-xi

Turn on the annotation overlays for pedagogical diagrams:

```
riemann-sum(
  x => x * x,
  domain: (0.0, 3.0),
  n: 6, method: "right",
  color: blue.lighten(80%),
  show-points: true,           // dots at evaluation points
  show-dx: true, dx-rect: 2,   // Δx bracket under rectangle 2
  show-xi: true,               // x₀ ... x₆ labels along the axis
)
```



7.13.3. xi-show-values

Set `xi-show-values: true` to display the actual numeric values of the subdivision points instead of the symbolic x_i notation:

```
riemann-sum(fn, domain: (0.0, 4.0), n: 4, method: "left",
            show-xi: true, xi-show-values: true)
```

7.13.4. Hatch patterns

Use `hatch` together with `hatch-spacing` and `hatch-stroke` to draw hatched rectangles. Set `color: none` for hatch-only (no fill):

```
riemann-sum(fn, domain: ..., n: 6, method: "mid",
            color: none,
            hatch: "ne", hatch-spacing: 4pt, hatch-stroke: blue + 0.6pt)
```

Available hatch styles: "ne", "nw", "h", "v", "cross", "grid".

7.14. volume-of-revolution - Volume of revolution diagram

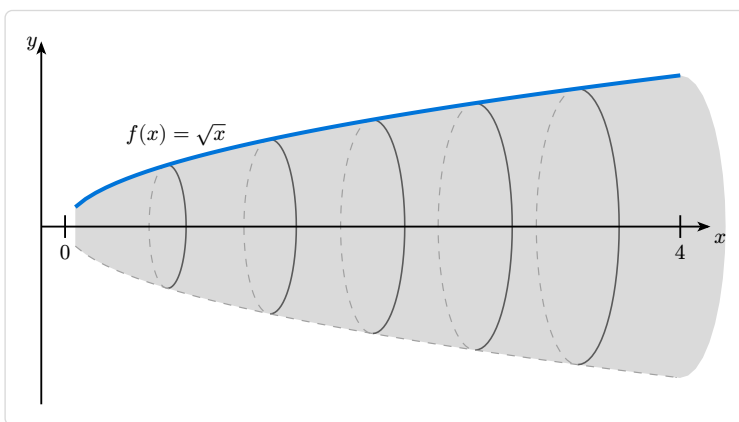
Draw a 3D-style solid generated by rotating a profile $y = f(x)$ around a horizontal or oblique axis. The renderer draws the filled body, front and back cap ellipses, intermediate disk cross-sections, axis arrow, coordinate y-axis, and optional labels.

7.14.1. Basic Example

Code

```
#volume-of-revolution(  
  x => calc.sqrt(x),  
  domain: (0.0, 4.0),  
  n-disks: 5,  
  width: 8.0, height: 4.0,  
  show-y-axis: true,  
  label-a: $0$, label-b: $4$,  
  label-f: $f(x)=\sqrt{x}$$,  
)
```

Preview

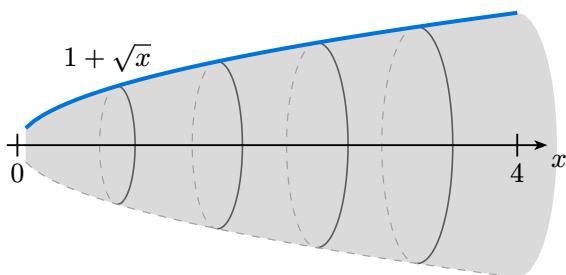


7.14.2. Axis of Revolution

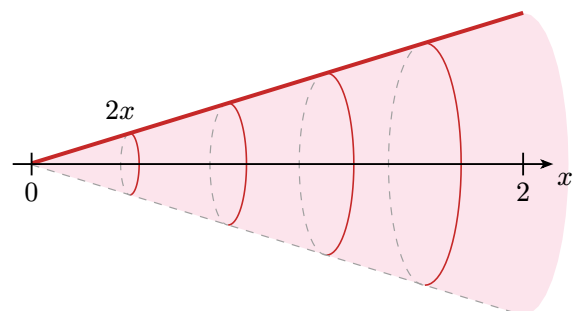
Use `axis-y` to revolve around $y = c$ (shifted horizontal axis), or `axis-slope` for an oblique axis $y = mx +$ `axis-y`:

```
// Around y = 1  
#volume-of-revolution(x => 1.0 + calc.sqrt(x), domain: (0, 4), axis-y: 1.0)  
  
// Around y = x (oblique)  
#volume-of-revolution(x => x * x, domain: (0.01, 2), axis-slope: 1.0)
```

Around $y = 1$



Around $y = x$ (oblique)



7.14.3. show-back and show-radius-marker

`show-back: false` hides the dashed back half and bottom profile — useful for side-by-side comparisons. `show-radius-marker: true` draws a vertical dimension marker on the profile (at `radius-marker-x`, default: the left cap):

```
#volume-of-revolution(fn, domain: ...,  
  show-back: false,  
  show-radius-marker: true,  
  radius-marker-x: 2.0, // marker at x=2 instead of the left cap  
  label-y:  $\sqrt{2}$ ,  
)
```

7.14.4. show-y-axis and y-axis placement

The coordinate y-axis is positioned automatically to the left of the solid by default. Use `y-axis-x` to fix it at a specific x-value, `y-axis-offset` to change the automatic gap, and `y-axis-extend` to control how far the axis extends above and below the solid:

```
#volume-of-revolution(fn, domain: ...,  
  show-y-axis: true,  
  y-axis-offset: 0.6, // more gap from the volume  
  y-axis-extend: (0.2, 0.5), // tighter below, more above  
)
```


8. Global Configuration

8.1. Setting Defaults

Use `set-plot-defaults` to configure defaults for all subsequent plots. Any `plot` parameter can be used, including `style`; unknown option names raise an error (typo protection):

```
#set-plot-defaults(  
  width: 8,  
  height: 6,  
  show-grid: "both",  
  minor-grid-step: 5,  
  style: (  
    axis: (stroke: navy + 0.8pt, arrow: (symbol: "stealth", fill: navy, scale: 0.55)),  
    ticks: (stroke: navy + 0.6pt, label-fill: navy),  
  ),  
)  
  
// All subsequent plots will use these defaults  
#plot(xmin: -3, xmax: 3, ymin: -2, ymax: 2, ...)
```

Per-call `style`: arguments are merged on top of the default style, so you can override individual properties without restating the full style.

8.2. Resetting Defaults

Reset to original defaults:

```
#reset-plot-defaults()
```

9. Styling

9.1. Custom Styles

Override default styles with the `style` parameter:

```
#plot(  
  xmin: -3, xmax: 3, ymin: -2, ymax: 2,  
  style: (  
    background: (fill: rgb("#202124")),  
    axis: (  
      stroke: white + 1pt,  
      arrow: (symbol: "stealth", fill: white, scale: 0.55),  
    ),  
    grid: (  
      major: (stroke: luma(120) + 0.6pt),  
      minor: (stroke: luma(80) + 0.3pt),  
    ),  
    ticks: (  
      length: 0.12,  
      stroke: white + 0.6pt,  
      label-size: 0.7em,  
      label-fill: white,  
    ),  
    labels: (  
      fill: white,  
    ),  
  ),  
  ...  
)
```

9.2. Default Style Values

Property	Default	Description
background.fill	none	Plot background fill
background.stroke	none	Plot background stroke
axis.stroke	black + 0.8pt	Axis line style
axis.arrow	(symbol: "stealth", fill: black, scale: 0.55)	Arrow head style
grid.major.stroke	luma(200) + 0.5pt	Major grid line style
grid.minor.stroke	luma(230) + 0.3pt	Minor grid line style
ticks.length	0.1	Tick mark length (cm)
ticks.stroke	black + 0.6pt	Tick mark style
ticks.label-size	0.65em	Tick label font size
ticks.label-fill	black	Tick label color
ticks.label-offset	0.15	Distance from tick to label
plot.stroke	blue + 1.2pt	Default function stroke
plot.samples	100	Default sample count
marker.size	0.12	Default marker size
labels.size	0.8em	Axis label font size

<code>labels.fill</code>	<code>black</code>	Axis label color
--------------------------	--------------------	------------------

10. Parameter Reference

10.1. plot Function

All parameters are named and optional; series specs are passed positionally. Unknown named arguments raise an error (typo protection).

Dimensions and Bounds:

Parameter	Type	Default	Description
width	length/float	6	Plot width — a length (2cm, 30mm, ...) or a number in cm
height	length/float	6	Plot height — a length or a number in cm
scale	float	1	Scale factor for entire plot
xmin	float	−5	Minimum x value
xmax	float	5	Maximum x value
ymin	float	−5	Minimum y value
ymax	float	5	Maximum y value

Axis Configuration:

Parameter	Type	Default	Description
xlabel	content	x	X-axis label
ylabel	content	y	Y-axis label
xlabel-pos	string/array	“end”	“end”, “center”, or (x, y)
ylabel-pos	string/array	“end”	“end”, “center”, or (x, y)
xlabel-anchor	string	“north”	Anchor for x label
ylabel-anchor	string	“east”	Anchor for y label
xlabel-offset	array	(0.0, −0.05)	X label offset (cm)
ylabel-offset	array	(-0.05, 0.0)	Y label offset (cm)
axis-x-pos	string/float/ none	0	“bottom”, “center”, a y-value, or none to hide the axis
axis-y-pos	string/float/ none	0	“left”, “center”, an x-value, or none to hide the axis
axis-x-extend	length/float/ array	(0pt, 0.3cm)	X-axis extension (left, right); lengths are absolute, bare numbers are data units (legacy)
axis-y-extend	length/float/ array	(0pt, 0.3cm)	Y-axis extension (bottom, top); lengths are absolute, bare numbers are data units (legacy)

Tick Configuration:

Parameter	Type	Default	Description
xtick	array/none	auto	Custom x tick positions

<code>ytick</code>	array/none	auto	Custom y tick positions
<code>xtick-step</code>	float	1	X tick spacing
<code>ytick-step</code>	float	1	Y tick spacing
<code>xtick-labels</code>	array/none	auto	Custom x tick labels
<code>ytick-labels</code>	array/none	auto	Custom y tick labels
<code>xtick-label-step</code>	int	1	Show x label every N ticks
<code>ytick-label-step</code>	int	1	Show y label every N ticks
<code>show-origin</code>	bool	true	Show “0” at origin
<code>origin-label-offset</code>	array	(-0.11, -0.11)	Origin “0” label offset from (0,0) in cm
<code>origin-label-anchor</code>	string	“north-east”	Origin “0” label anchor
<code>origin-leader</code>	bool	true	Draw a subtle leader line from origin label toward (0,0)
<code>origin-leader-stroke</code>	stroke	black + 0.6pt	Origin leader stroke
<code>origin-leader-gap</code>	float	0.025	Gap from (0,0) before leader starts (cm)
<code>origin-leader-end-gap</code>	float	0.025	Gap before the label anchor (cm)
<code>unit-label-only</code>	bool	false	Show only “1” on axes
<code>tick-label-size</code>	length	0.65em	Tick label font size
<code>axis-label-size</code>	length	0.8em	Axis label font size
<code>min-tick-spacing</code>	float	0.4	Minimum distance between ticks (cm); auto-widens step to the next “nice” value (1, 2, 5, 10...) when the range is large
<code>hide-crossed-tick-labels</code>	bool	true	Hide individual tick labels that a plotted curve crosses, keeping labels legible
<code>show-end-ticks</code>	bool	true	Keep the tick/label at <code>xmax/ymax</code> when it lands on a tick (e.g. show “5”), pushing the axis slightly past it so the arrow clears the label

Grid Configuration:

Parameter	Type	Default	Description
show-grid	bool/string	false	true, false, “major”, “minor”, “both”
minor-grid-step	int	5	Subdivisions per major tick
grid-label-break	bool	true	Break grid lines around labels

Styling:

Parameter	Type	Default	Description
style	dictionary	none	Style overrides (see Styling section)
font	str/array	auto	Font applied to all plot-generated text (tick labels, axis labels, origin, annotations); auto = document font
series	array	none	Pre-built array of series specs (alternative to positional args)

10.2. Function/Data Specification

Each plot item is a dictionary with these fields:

Field	Type	Description
fn	function	Function to plot: $x \Rightarrow y$ (return none to create a hole/discontinuity)
data	array	Data points: $((x_1, y_1), (x_2, y_2), \dots)$
connect	bool	Connect data points with a line (default: true for fn , false for data)
domain	array	Function domain: $(x_{\min}, x_{\max})$
samples	int	Number of samples for function (default: 100)
stroke	stroke	Line style (default: blue + 1.2pt)
mark	string	Marker type (default: “none”)
mark-size	float	Marker size in cm (default: 0.12)
mark-fill	color	Marker fill color (default: black)
mark-stroke	stroke	Marker stroke style (default: black + 0.8pt)
label	content	Label text placed near the curve
label-pos	float	Position along the curve in $[0, 1]$ (default: 1.0)
label-side	string	Label placement relative to curve: “above”, “below”, “left”, “right”, “above-left”, etc.
label-anchor	string	CeTZ anchor override (alternative to label-side)

10.3. riemann-sum Function

Parameter	Type	Default	Description
fn	function	—	Function $f(x)$ to approximate (positional or fn:)
domain	array	plot range	Interval (a , b); overrides the plot's x-range
n	int	4	Number of rectangles
method	string	"right"	"left", "right", "mid", "lower", "upper"
baseline	float	0.0	Y-level of rectangle bases (default: x-axis)
color	color	luma(220)	Rectangle fill; none for hatch-only
stroke	stroke	luma(80) + 0.6pt	Rectangle border stroke
hatch	str/dict/none	none	Hatch pattern: "ne", "nw", "h", "v", "cross", "grid" — or (style: "ne", spacing: 5pt, stroke: red + 1pt)
hatch-spacing	length	5pt	Spacing between hatch lines
hatch-stroke	stroke	luma(80) + 0.5pt	Stroke for hatch lines
samples	int	20	Samples per sub-interval for "lower" / "upper" methods
show-points	bool	false	Draw a dot at each evaluation point
point-color	color	rgb("#c94a00")	Dot fill color
point-size	float	0.07	Dot radius in cm
point-label	content/auto/none	none	Arrow label near dots; auto = method name, none = dots only
point-label-pos	array/auto	auto	(x, y) in data coords; auto = upper-right of dots
show-dx	bool	false	Draw a Δx dimension bracket under one rectangle
dx-rect	int/auto	auto	Rectangle to annotate (0-based); auto = middle rectangle
dx-label	content	Δx	Label displayed inside the bracket
show-xi	bool	false	Draw $x_0, x_1, \dots, x_n$ at subdivision points
xi-labels	array/auto	auto	Custom label array; auto = subscripted x_i notation
xi-show-values	bool	false	Show numeric x-values instead of x_i notation

10.4. volume-of-revolution Function

solid-of-revolution is an alias kept for backward compatibility.

Parameter	Type	Default	Description
fn	function	—	Profile function $f(x)$; the curve that is revolved
domain	array	(0.0, 4.0)	Interval (a, b) — the revolution interval
n-disks	int	4	Number of intermediate disk cross-sections to show
width / height	length/float	5.0 / 3.5	Canvas size — a length or a number in cm
samples	int	60	Number of points sampled along the profile curve
axis-y	float	0.0	Y-value of horizontal revolution axis ($y = c$)
axis-slope	float	0.0	Slope m : revolution axis is $y = mx + \text{axis-y}$
show-axis	bool	true	Draw the revolution axis with an arrowhead
show-y-axis	bool	false	Draw a coordinate y-axis at the left cap (<code>show-yaxis</code> is a deprecated alias)
y-axis-x	float/auto	auto	X-position for the y-axis; <code>auto</code> = left of volume
y-axis-offset	float	0.45	Gap between y-axis and the volume when <code>auto</code>
y-axis-extend	array	(0.35, 0.45)	Y-axis padding (below, above) the solid
show-radius-marker	bool	false	Draw a vertical radius dimension marker at <code>radius-marker-x</code>
radius-marker-x	float/auto	auto	X-position of the radius marker; <code>auto</code> = left cap (<code>yaxis-x</code> is a deprecated alias)
show-back	bool	true	Show the back half of the solid (dashed ellipses + bottom curve)
show-labels	bool	true	Show a , b , f labels
profile-stroke	stroke	blue + 1.5pt	Top profile curve stroke
disk-color	color	luma(218)	Solid body fill color
disk-stroke	stroke	luma(90) + 0.6pt	Disk edge stroke
axis-stroke	stroke	black + 0.7pt	Revolution axis stroke
label-a / label-b	content	\$a\$ / \$b\$	Labels at domain endpoints

label-f	content	f	Function label near the profile curve
label-y	content	y	Label for the coordinate y-axis

10.5. zoom - Spy / zoom inset

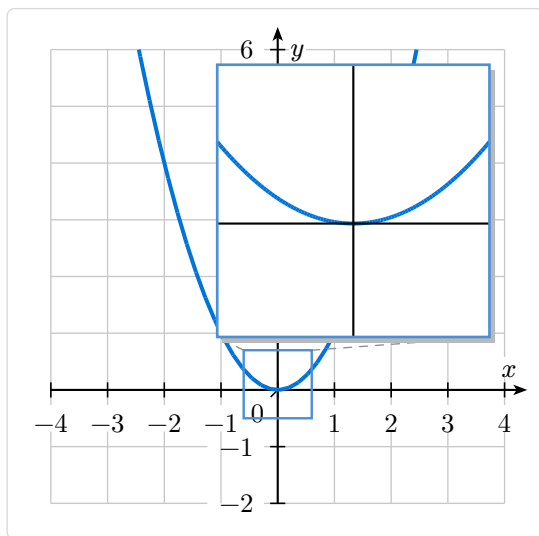
Add a magnified sub-view (spy glass) of any region of the main plot. The `zoom()` helper returns a series spec that is passed directly to `plot()`. It draws a highlighted region on the main plot and a zoomed inset box with connector lines. All series types are re-rendered inside the inset — curves, points, fills, Riemann rectangles, reference lines and parametric curves — clipped to the lens shape (annotations and labels are not repeated).

10.5.1. Basic Example

Code

```
#plot(  
  xmin: -4, xmax: 4, ymin: -2, ymax: 6,  
  show-grid: "major",  
  (fn: x => x * x, stroke: blue + 1.5pt),  
  zoom(center: (0, 0.1), size: 0.9, magnification: 4, at: "top-right"),  
)
```

Preview

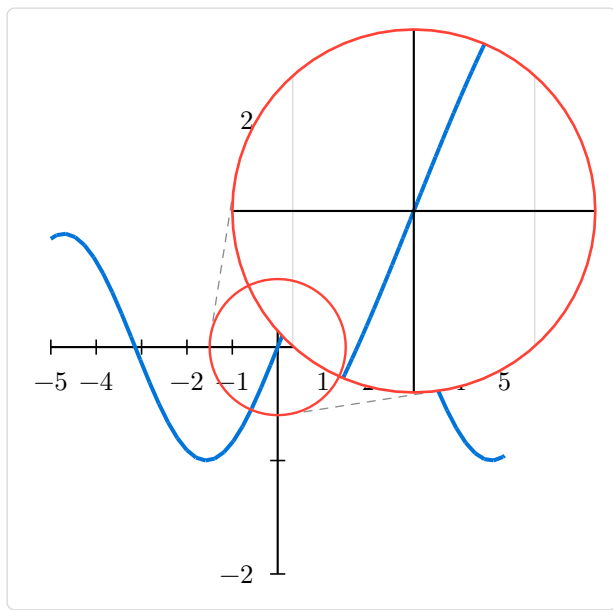


10.5.2. Circular lens, explicit region

Code

```
#plot(
  xmin: -5, xmax: 5, ymin: -2, ymax: 2,
  (fn: x => calc.sin(x), stroke: blue + 1.5pt),
  zoom(region: (-0.5, -0.6, 0.5, 0.6),
    lens-shape: "circle", magnification: 5,
    at: (3, 1.2), accent: red),
)
```

Preview



10.5.3. Parameters

Parameter	Type	Default	Description
region	array	auto	Spy glass corners (x1, y1, x2, y2) in data coords. Either region or center + size is required.
center	array	auto	Spy glass center (cx, cy) in data coords (use with size)
size	length/float	auto	Spy glass size (length, or number in cm) — ensures a square glass regardless of axis ratio
at	array/str	auto	Inset box center: (x,y) data coords, or a placement keyword: "top-right", "top-left", "bottom-right", "bottom-left", "top", "bottom", "left", "right". auto = smart opposite-quadrant placement
width / height	length/float	auto	Inset box size (length, or number in cm). auto = derived from magnification or 3.5
magnification	float	auto	Zoom factor — inset size = spy glass canvas size × factor
lens-shape	str	"rect"	"rect" or "circle"
connect	bool	true	Draw connector lines between spy glass and inset
accent	color	blue	Accent color for borders and region highlight
region-fill	color	auto	Fill tint inside the spy glass region
region-stroke	stroke	auto	Border of the spy glass region
box-fill	color	white	Background fill of the inset box
box-stroke	stroke	auto	Border of the inset box

connector-stroke	stroke	auto	Dashed gray connector lines by default
connector-fill	color	none	Fill the trapezoid between connector lines
shadow	bool	auto	Drop shadow behind the inset (default true for rect with fill)
show-inset-grid	bool	true	Subtle grid inside the inset aligned to main ticks
show-magnification	bool	false	Show $\times N$ label in inset corner
label	content	none	Custom label inside the inset (overrides the $\times N$ label)

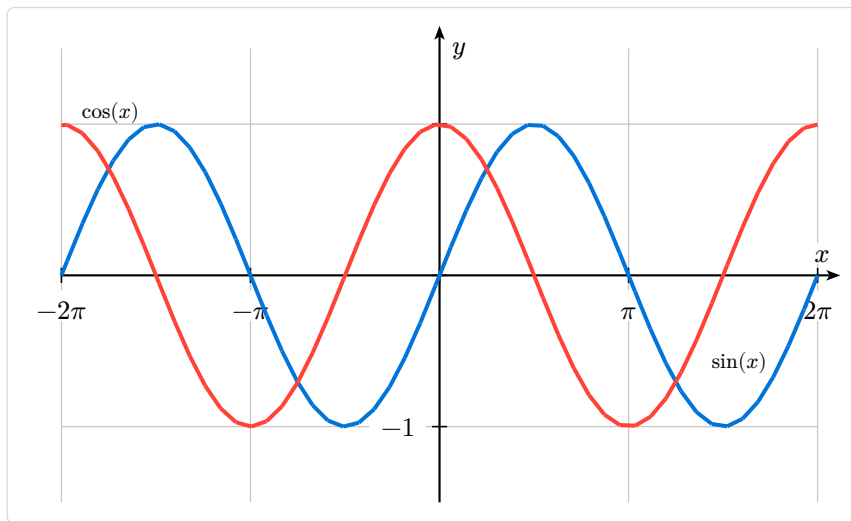
11. Complete Examples

11.1. Trigonometric Functions

Code

```
#plot(  
  width: 10, height: 6,  
  xmin: -2 * calc.pi, xmax: 2 * calc.pi,  
  ymin: -1.5, ymax: 1.5,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  xtick: (-2*calc.pi, -calc.pi, 0, calc.pi, 2*calc.pi),  
  xtick-labels: ( $-2\pi$ ,  $-\pi$ ,  $0$ ,  $\pi$ ,  $2\pi$ ),  
  show-grid: "major",  
  (fn: x => calc.sin(x), stroke: blue + 1.5pt, label:  $\sin(x)$ , label-pos: 1),  
  (fn: x => calc.cos(x), stroke: red + 1.5pt, label:  $\cos(x)$ , label-pos: 0),  
)
```

Preview

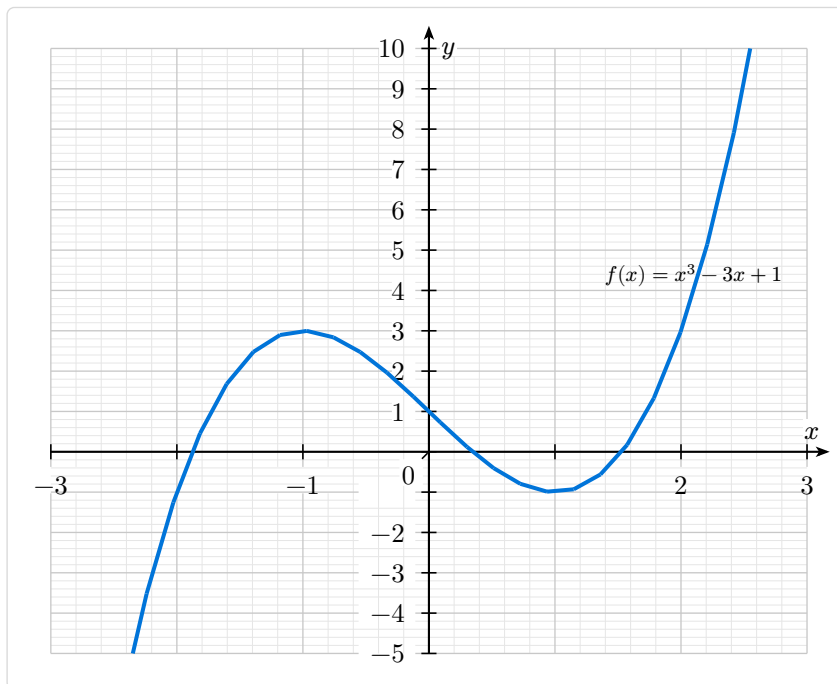


11.2. Polynomial with Fine Grid

Code

```
#plot(  
  width: 10, height: 8,  
  xmin: -3, xmax: 3, ymin: -5, ymax: 10,  
  xlabel: $x$, ylabel: $y$,  
  show-grid: "both",  
  minor-grid-step: 5,  
  (  
    fn: x => x * x * x - 3 * x + 1,  
    stroke: blue + 1.5pt,  
    label: $f(x) = x^3 - 3x + 1$,  
    label-side: "above",  
    label-pos: 0.85,  
  ),  
)
```

Preview

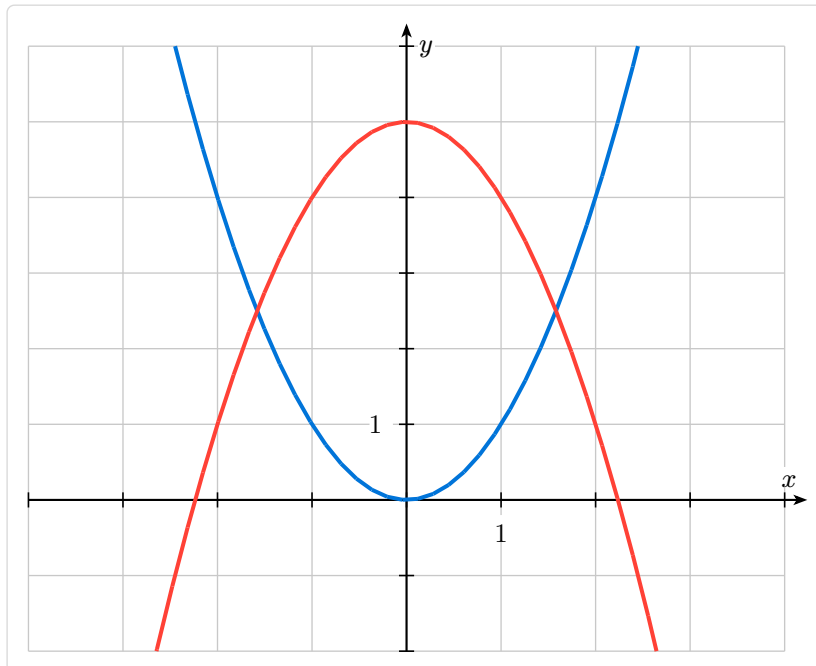


11.3. Minimal Style Plot

Code

```
#plot(  
  width: 10, height: 8,  
  xmin: -4, xmax: 4, ymin: -2, ymax: 6,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "major",  
  unit-label-only: true,  
  show-origin: false,  
  (fn: x => x * x, stroke: blue + 1.5pt),  
  (fn: x => -x * x + 5, stroke: red + 1.5pt),  
)
```

Preview



11.4. Data with Trend Line

Code

```
#plot(
  width: 10, height: 7,
  xmin: 0, xmax: 6, ymin: 0, ymax: 12,
  xlabel: "Time (s)", ylabel: "Distance (m)",
  axis-x-pos: "bottom", axis-y-pos: "left",
  show-grid: "both",
  minor-grid-step: 5,
  (fn: x => 2 * x, stroke: gray + 1pt, domain: (0, 6)), // Trend line
  (
    data: ((0.5, 1.2), (1, 2.3), (2, 3.8), (3, 6.2), (4, 7.9), (5, 10.1)),
    mark: "o",
    mark-size: 0.12,
    stroke: none,
  ),
)
```

Preview

